Kunaal Shivakumar

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ACADEMIC QUALIFICATION

Bachelor of Engineering in **Computer Science**, BMS Institute of Technology and Management (BMSIT&M), Bengaluru; Visvesvaraya Technological University (VTU); CGPA: 8.47/10; Jun'24

WORK EXPERIENCE

Associate Software Developer in Test*, Q2 Software, Bengaluru** ***Aug'24 – Present

- Designing and maintaining a scalable regression automation suite using Python, Selenium, and the unittest framework, automating 100+ critical test cases to ensure software reliability for major financial institutions.
- Integrating the automated regression suite into Jenkins CI/CD pipelines, enabling continuous testing and delivering real-time feedback to developers on every code commit.
- Developing and executing an API automation suite for smoke testing using Postman, achieving over 95% endpoint coverage and ensuring the reliability and integrity of backend services
- Performing comprehensive manual, functional, and database testing for large-scale, multi-zone projects, 8+ financial institutions, leveraging BrowserStack for cross-platform validation, Postman for API testing, and Microsoft SQL Server for backend data verification
- Contributing across the full Agile lifecycle, including sprint planning, backlog grooming, and Go-Live testing, to ensure a clear understanding of requirements and successful deployments

INTERNSHIP

QA Intern*, Q2 Software, Bengaluru** ***Apr'24 – Aug'24

- Diagnosed and resolved 100+ failures in automated smoke test suites within Jenkins, reducing the overall failure rate by 85% and significantly enhancing the stability and reliability of the CI pipeline
- Executed a comprehensive suite of 200+ combined manual smoke and regression test cases for each financial client across both desktop and mobile platforms to ensure end-to-end application quality and performance
- Ensured high-quality software releases for critical banking applications by executing robust manual and automated smoke/regression test suites using Jenkins, BrowserStack, and Postman, contributing to improved end-user experience for both retail and commercial customers

ACADEMIC PROJECTS

Title: Bone Tumour Detection and Classification Using Deep Learning Based on Computed Tomography
Duration: Aug– May'24
Team size: 4
Summary: Developed a deep learning-based system for automated bone tumour detection from CT scans, leveraging CNN and transfer learning. Fine-tuned CNN, VGG-16 and DenseNet121 through hyperparameter optimisation, achieving a 91.3% accuracy and an F1-score of 0.89 with DenseNet121, marking a 4.3% improvement over VGG-16. Preprocessed 6,000+ CT images using normalisation, resizing, and data augmentation to enhance model robustness. Deployed the model as a Flask-based web app for real-time classification and validated its clinical viability through comprehensive performance analysis
Individual role: Handled data preprocessing and model training using TensorFlow and Keras, fine-tuning CNN, VGG-16 and DenseNet121. Optimised performance on 6,000+ CT images and built the frontend for a Flask web app, enabling real-time tumour classification.

Title: Online Tourism Management System
Duration: Nov'22 – Jan'23

Team size: 2
Description: Designed and developed a dynamic tourism website using HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL. Integrating secure user authentication with PHP session handling and database validation. Built a backend admin dashboard with full CRUD functionality for managing tour packages and user data. Implemented interactive tour package listings with detailed itineraries, pricing, and availability to streamline bookings. Emphasised clean UI/UX and ensured secure, high-performance backend operations.
Individual Role: Built the responsive frontend using HTML, CSS, JavaScript, and Bootstrap with a focus on intuitive UX. Designed the MySQL database schema and optimised SQL queries for efficient data handling and full CRUD functionality.

PUBLICATIONS

- *Kunaal Shivakumar, co-author, "Bone Tumour Detection and Classification Using Deep Learning Based on Computed Tomography,"* published in IEEE Xplore Digital Library (DOI: 10.1109/NMITCON62075.2024.10699180), Oct'24

PRESENTATIONS

- Presented a paper on 'Decoding DeFi: A Comparative Analysis of Popular and High-Traffic Decentralized Finance Applications' at the International Conference on Networks, Multimedia and Information Technology (NMITCON-2025), Bengaluru, Aug'25
- Presented a paper on 'Bone Tumour Detection and Classification Using Deep Learning Based on Computed Tomography' at the International Conference on Networks, Multimedia and Information Technology (NMITCON-2024), Bengaluru, Aug'24

CERTIFICATIONS

- Python Automation and Testing in LinkedIn Learning, Oct'24
- Jenkins Essential Training in LinkedIn Learning, Jun'24
- Postman Essential Training in LinkedIn Learning, May'24
- Python Essential Training in LinkedIn Learning, Apr'24

TECHNICAL SKILLS

- Languages: C, Java, Python
- Web Technologies: HTML5, CSS3, PHP, JavaScript, Flask
- Database: MySQL
- Others: Selenium, Jenkins, Postman, TensorFlow

EXTRACURRICULAR ACTIVITIES

- Achieved 4th position in Visvesvaraya Technological University State-Level Inter-Collegiate Cricket Tournament, Feb'24
- Won Visvesvaraya Technological University North Zone Inter-Collegiate Cricket Tournament, Jan'24
- Served as a Student Placement Co-ordinator coordinating campus placement activities (recruitment drives, pre-placement sessions, interviews) and assisting students with resume development and mock interviews, Mar'23 – Jun'24
- Organised and managed a wide range of inter-college sports events and tournaments, overseeing logistics, scheduling, and event execution while collaborating with faculty, sports committees, and external institutions as a Student Sports Co-ordinator, Jun'23